MODIFIED PROJECT

Finding the Shortest Bus Route Between a Source and Destination

Problem Statement

Passengers need to find the best route between a chosen starting bus stop and a specific destination. Reporting travel times to all stops does not directly show the route selected by a passenger. The system should calculate the minimum travel time and display the sequence of stops between the chosen source and destination.

Solution

Represent the bus network as a weighted graph, where vertices are stops and edge weights are travel times. Ask the user for source and destination stops and validate them. Apply Dijkstra's algorithm from the source, storing each stop's predecessor when a shorter route is found. Reconstruct and display the shortest route and its total travel time. If the destination is unreachable, display a message that no route is available.